

CounterHelp: Promoting Online Civil Courage Among Young People Through AI-Generated Counter Speech

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Abstract

We present *CounterHelp*, a mobile app designed to support adolescents and young people in generating customised counter speech in response to hateful comments on TikTok. CounterHelp allows users to specify counter strategies to combat hate speech. After users share a hateful comment with CounterHelp, it retrieves TikTok metadata to capture its context. Leveraging large language models, CounterHelp generates customised and context-sensitive counter speech in one of four predefined styles, particularly tailored for adolescents. A user experience lab study confirms the effectiveness and usability of CounterHelp and the generated counter speech.

CCS Concepts

• Human-centered computing → Social media; • Computing methodologies → Natural language generation.

Keywords

Online Hate, Large Language Model, Social Media, TikTok

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1 Introduction

Hate speech is widespread on social media platforms, aiming to insult, marginalise, or incite violence against individuals or groups based on personal characteristics, e.g., gender, religion, or political beliefs [2]. This phenomenon is partly driven by the anonymous and unmoderated nature of social media platforms, which enables users to express opinions without considering the potential harm they may cause. Hate speech can lead to online conflict and seriously impact the psychological well-being of its targets [1, 8], while its

influence can spread across social media and may impact numerous other users [4]. Adolescents often struggle to respond appropriately to online hate and need more attention and external guidance to protect themselves from potential harm [3, 11, 13].

Counter speech is a direct and constructive response intended to mitigate the effects of hate speech [2, 6]. It has been studied and advised as an effective tool to empower young and adolescent people in actively challenging online hate and fostering empathy for victims [12]. The purpose of counter speech is not only to respond to harmful content, but also to halt the spread of hateful messages, support victims, and educate silent bystanders to promote positive values. Citizen-driven counter speech [2] is increasingly seen as a promising alternative to content removal by platform moderators. Successful counter speech relies on selecting suitable strategies, such as amplification, education, humour, or shaming, which have been shown to be effective in various contexts [7]. To be effective, counter speech must be context-aware. On short-form video platforms like TikTok, it should take into account content, style, and tone of the related video and the communication thread leading to the hate speech. Furthermore, the style of counter speech should be tailored to the target group. This aspect is often neglected in previous research but presents a special focus of our work.

Contribution. We demonstrate *CounterHelp*¹, a mobile app that assists young TikTok users in generating context-sensitive counter speech. CounterHelp is designed and developed in a joint effort of computer scientists, sociologists, and adolescents to meet the needs of our target group. It encourages adolescents to actively combat online hate, fosters self-reflection, and aims to mitigate the harmful effects of hateful content.

Related work. Only a few related efforts exist in this domain. An example is “Schneller Konter”², which enables users to create visual and textual counter speech from a predefined set of templates. Its suggestions are generic and do not account for a specific hate comment, its context, and audience. “Detect Then Act”³ provides supporting material for counter speakers. We build upon previous attempts towards automated counter speech generation, such as CounterGeDi [9]. Similarly to CounterGeDi, CounterHelp leverages recent advancements in large language models (LLMs) to provide context-sensitive counter speech. It goes beyond the state of the art by adding customisation to the generation process and enabling seamless integration into the targeted social media platform.

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¹Supplemental video: <https://phaidra.fhstp.ac.at/o:7289>, last visited 07/2025

²<https://www.schnellerkonter.at>, last visited 07/2025

³<https://dtct.eu>, last visited 07/2025

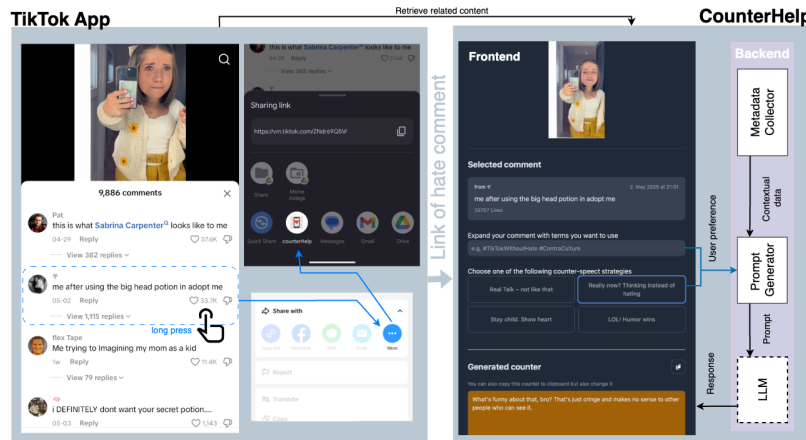


Figure 1: Workflow and design of CounterHelp.

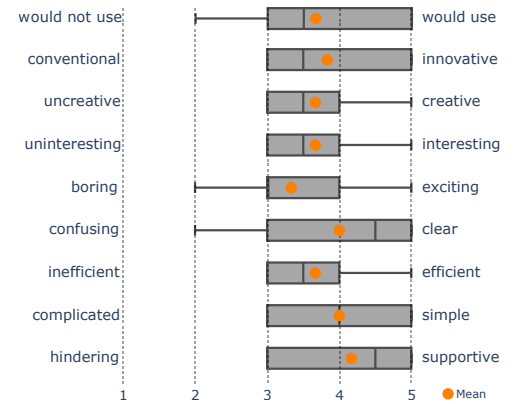


Figure 2: User satisfaction survey.

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2 Design of the CounterHelp Application

CounterHelp is implemented as a Progressive Web App (PWA), making it usable on both iOS and Android smartphones. The overall framework is depicted in Figure 1. As users browse TikTok through their native app, they can send a hateful comment to CounterHelp by long-pressing the comment, choosing the ‘share’ button in the pop-up menu, and selecting CounterHelp. This opens the CounterHelp interface, which receives the hyperlink to the hateful comment. The CounterHelp interface consists of three sections: At the top, the app displays the video and the hateful comment (provided via the hyperlink). In the center, users can customize the generated counter speech in two ways: a free-text input for keywords (e.g., hashtags and emojis) to be included in the counter speech, and a selection menu for the overall style of the counter speech. Users can choose from four counter speech styles: *education*, *empathy*, *warning of consequences*, and *humour and sarcasm* [7]. At the bottom, the generated counter speech is displayed. We renamed the four styles to better suit adolescents, e.g., renaming empathy to “Stay kind. Show heart”.

On the server side, we implemented three submodules: a meta-data collector, a prompt generator, and an LLM module. To capture the context of the hateful comment, the *metadata collector* fetches the video’s title, captions, and all preceding comments. The comments are organised as a tree to capture their hierarchical relations. The *prompt generator* fills a predefined prompt template (specific for each counter speech style) with relevant information from the collected metadata and user-provided keywords. The *LLM module* takes the customized prompt and generates a counter speech comment using either a local model (Gemma 3n, 4B) or the GPT-4o mini API. Our backend server sends the generated counter speech to the frontend, where users can edit it or copy it directly for posting.

3 Prompt Engineering

The quality of LLM responses depends to a large extent on the design of the prompt. In our prompt engineering process, we faced several challenges in ensuring that the generated counter speech exhibits all desired properties and is well-tailored to adolescents. The following three aspects were most important. First, we observe that hateful TikTok comments are often very short, providing limited context, making it hard to understand the topic behind the hateful message. To help the LLM identify the right topic, we provided as context the direct parent comment of the hateful message as well as the global parent comment (i.e., first comment in the conversation thread). Second, a key challenge is formulating counter speech in a style that resonates with young people. We use the role prompting method [5], which assigns a specific role or persona to guide the style of the response. In addition, we explicitly provide a curated list of keywords suggested by sociologists intensively working together with adolescents, which are commonly used by young users. Third, we observe that counter speech generated using the same style often exhibits repetitive patterns, both in terms of wording and semantic reasoning. To enhance the diversity of responses, we introduce variability by randomly altering aspects of the assigned persona (e.g., age and gender). In addition, we adjust the temperature parameter, which increases the variation in the resulting counter speech recommendations.

4 User Experience Study

We conducted a lab study to gather feedback on the usability of CounterHelp and the quality of the generated counter speech. Nine participants completed the preliminary survey. The results of the *User Experience Questionnaire* (UEQ-S) [10], adapted for the target group, indicate that users generally rated the system positively across all attributes, with mean scores leaning toward the favourable end of the scale (Figure 2). This suggests that the usability of CounterHelp is high. In addition, eight of nine participants stated that they would use CounterHelp and a majority of participants assessed the generated counter speech as sounding natural and human.

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